

RUDRA SHUKLA

rudrashukla.ca | github.com/rudrashukla7 | +1 (647)-654-9116 | rudra.shukla@uwaterloo.ca | in/shukla-rudra

EDUCATION

University of Waterloo

Waterloo, ON

Candidate for B.A.Sc in Mechatronics Engineering

September 2025 – Present

- **Relevant Coursework:** Data Structure and Algorithms, Engineering Graphics and Design, Circuits, Statics

PROJECTS

Toyota Innovation Challenge (Safety Award Winner)

June 2025

Python, OpenCV, Robotic Arm, Voice Recognition, Raspberry Pi, UI Development

Waterloo, ON

- Developed an intelligent manufacturing assistant integrating **computer vision**, **robotic arm control**, and voice-activated workflows to improve operational efficiency and workplace safety.
- Implemented **real-time defect detection**, **bin localization**, and a **voice-controlled interface**, earning the team the Safety Award at the Toyota Innovation Challenge for enhancing operator awareness and industrial safety.

AutoniMake - AI Robotics Platform

March 2026

Python, C/C++, OpenCV, TensorFlow, Raspberry Pi, ESP32, Embedded Systems, UI Development

Waterloo, ON

- Built a code-free AI robotics platform enabling users to collect image data, **train Convolutional Neural Network (CNN) models**, and deploy real-time predictions through a web interface.
- Developed a real-time vision pipeline, and designed a distributed system where a Raspberry Pi handles inference and communicates with ESP32 devices with a custom **UART protocol** for low-latency hardware control.

Automatic Blinds Controller

April 2026

ESP32, C/C++, Embedded Systems, IoT Web Server, 3D Printing, Circuit Design

Ajax, ON

- Built an ESP32-based automatic blinds controller enabling **precise bidirectional stepper motor control**.
- Developed a WiFi-enabled web interface with real-time control (manual, timed, stop), and **implemented dual-core tasking**, microstepping, and power regulation for smooth, reliable operation.

EXPERIENCE

Firmware & Perception Core Member

May 2026 – Present

WATonomous (Humanoid Team)

Waterloo, ON

- Implementing **SLAM**-based perception pipelines for humanoid **robot localization and mapping**.
- Developing embedded **C++ firmware** for custom motor control, regulation, and **real-time actuation**.
- Designing **hardware-software interfaces** and communication protocols for robotic subsystems.

Software Quality Analyst

January 2026 – April 2026

i4i (Infrastructures for Information Inc.)

Toronto, ON

- Collaborated with developers throughout the **Software Development Life Cycle (SDLC)** to track, retest, and validate bug fixes, ensuring alignment with functional requirements and release readiness.
- Executed structured test plans and **analyzed system behavior** across normal and edge-case conditions to identify defects and improve software stability, reliability, and production performance.

Robotics Instructor

January 2023 – August 2025

OBOTZ - IQ Brainers Academy

Ajax, ON

- Designed and delivered a hands-on robotics curriculum for **20+ students** using Arduino-based projects.
- Collaborated with industry professionals to **integrate real-world robotics** concepts into STEM programs.

Mechanical and Design Lead - FIRST Tech Challenge

September 2021 – June 2025

J. Clarke Richardson Collegiate

Ajax, ON

- Designed and built robot components using **SolidWorks**, and led development of **drivetrain and intake systems**.
- Collaborated with the software team to integrate autonomous functions, **improving scoring consistency by 75%**.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, HTML/CSS, JavaScript

Hardware: ESP32, Arduinos, Circuit Design, Raspberry Pi, Soldering

Tools: OpenCV, PLC, 3D Printing, SketchUp, Linux, Microsoft Office, FoxGlove, Pytorch, AutoCAD, SolidWorks.